Sample Paper 13

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

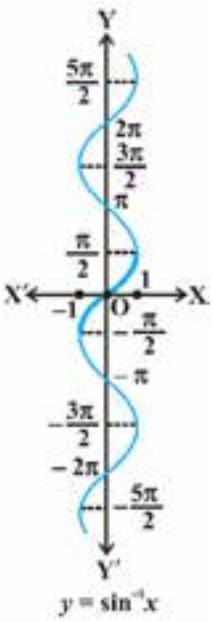
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. What is the value of y when $x = 1$ in the graph of $y = \cos^{-1}x$?



(a) 0 (b) $\pi/2$

(c) 1 (d) π

2. If $\int_0^a f(2a-x)\,dx = m$ and $\int_0^a f(x)\,dx = n$, then $\int_0^{2a} f(x)\,dx$ is equal to

(a) $2m+n$

(b) $m+2n$

(c) $m-n$

(d) $m+n$

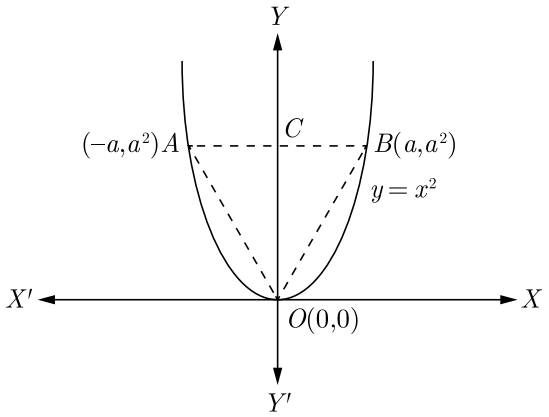
3. The area bounded by $y = |\sin x|$, X-axis and the lines $|x| = \pi$ is

(a) 2 sq units (b) 3 sq units

(c) 4 sq units (d) None of these

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4. The given figure shows a ΔAOB and the parabola $y = x^2$. The ratio of the area of the ΔAOB to the area of the region AOB of the parabola $y = x^2$ is equal to



(a) $\frac{3}{5}$ (b) $\frac{3}{4}$

(c) $\frac{7}{8}$ (d) $\frac{5}{6}$

5. The angle between the lines $x = 1$, $y = 2$ and $y = -1$, $z = 0$ is

(a) 30° (b) 60°

(c) 90° (d) 0°

6. Order of the equation $\left(1 + 5\frac{dy}{dx}\right)^{3/2} = 10\frac{d^3y}{dx^3}$ is

(a) 2 (b) 3

(c) 1 (d) 0

7. If x is real, then minimum value of $x^2 - 8x + 17$ is

(a) -1 (b) 0

(c) 1 (d) 2

8. If $P(A) = 0.5$, $P(B) = 0.4$ and $P(A \cap B) = 0.3$, then $P\left(\frac{A'}{B}\right)$ is equal to

(a) $\frac{1}{3}$ (b) $\frac{1}{2}$

(c) $\frac{2}{3}$ (d) $\frac{3}{4}$

9. If $\hat{a}, \hat{b}$ and $\hat{c}$ are unit vectors such that $\vec{a} + \vec{b} + \vec{c} = 0$. Then, which one of the following is correct?

(a) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a} = 0$

(b) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a} \neq 0$

(c) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{a} \times \vec{c} = 0$

(d) $\vec{a} \times \vec{b}, \vec{b} \times \vec{c}, \vec{c} \times \vec{a}$ are mutually perpendicular.

10. The line joining the points $(1,1,2)$ and $(3,-2,1)$ meets the planes $3x + 2y + z = 6$ at the point

(a) $(1,1,2)$ (b) $(3,-2,1)$

(c) $(2,-3,1)$ (d) $(3,2,1)$

11. If $f(x) = \begin{cases} \frac{3\sin \pi x}{5x}, & x \neq 0 \\ 2k, & x = 0 \end{cases}$ is continuous at $x = 0$, then the value of k is

(a) $\frac{\pi}{10}$ (b) $\frac{3\pi}{10}$

(c) $\frac{3\pi}{2}$ (d) $\frac{3\pi}{5}$

12. For two events A and B, if $P(A) = P\left(\frac{A}{B}\right) = \frac{1}{4}$ and $P\left(\frac{B}{A}\right) = \frac{1}{2}$, then

(a) A and B are independent events

(b) $P\left(\frac{A'}{B}\right) = \frac{3}{4}$

(c) $P\left(\frac{B'}{A}\right) = \frac{1}{2}$

(d) All of the above

13. The projection of $\vec{a} = 3\hat{i} - \hat{j} + 5\hat{k}$ on $\vec{b} = 2\hat{i} + 3\hat{j} + \hat{k}$ is

(a) $\frac{8}{\sqrt{35}}$ (b) $\frac{8}{\sqrt{39}}$

(c) $\frac{8}{\sqrt{14}}$ (d) $\sqrt{14}$

14. If $A = \begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}$ and A^2 is the identity matrix, then x is equal to

(a) -1 (b) 0

(c) 1 (d) 2

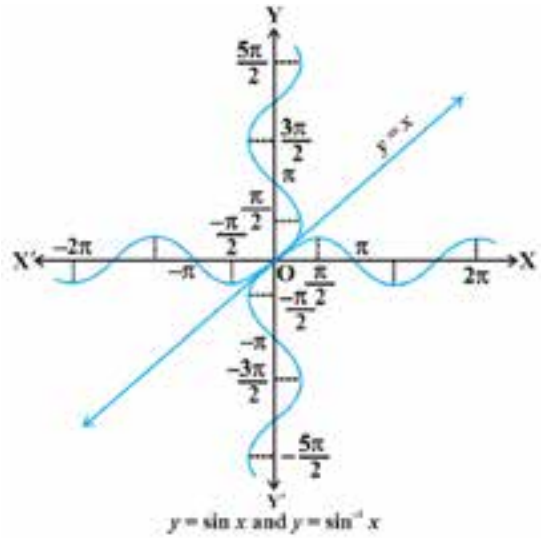
15. $\int_0^{\pi/2} \frac{\sin x - \cos x}{1 + \sin x \cos x}\, dx$ is equal to

(a) 0 (b) $\frac{\pi}{4}$

(c) $\frac{\pi}{2}$ (d) π

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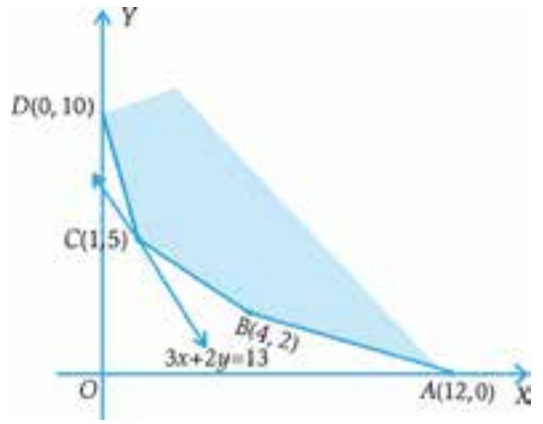
16. What is the value of y when $x = -1$ in the graph of $y = \text{cosec}^{-1} x$?



(a) 0 (b) $\pi/2$

(c) $-\pi/2$ (d) $-\pi/2$ or $\pi/2$

17. With the same feasible region, find the minimum value of $z = 4x + y$.



(a) 48 (b) 18

(c) 9 (d) 10

18. $3a\int_0^1 \left(\frac{ax-1}{a-1}\right)^2 dx$ is equal to

(a) $a - 1 + (a-1)^{-2}$ (b) $a + a^{-2}$

(c) $a - a^2$ (d) $a^2 + \frac{1}{a^2}$

19. **Assertion :** The matrix $A = \frac{1}{3}\begin{bmatrix} 1 & -2 & 2 \\ -2 & 1 & 2 \\ -2 & -2 & -1 \end{bmatrix}$ is an orthogonal matrix.

Reason : If A and B are orthogonal, then AB is also orthogonal.

(a) Assertion is true, reason is true, reason is a correct explanation for assertion.

(b) Assertion is true,reason is true, reason is not a correct explanation for assertion.

(c) Assertion is true, reason is false.

(d) Assertion is false, reason is true.

20. **Assertion :** If $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$, then $\text{adj}(\text{adj}\, A) = A$.

Reason : $|\text{adj}(\text{adj}\, A)| = |A|^{(n-1)^2}$, where A be n rowed non-singular matrix.

(a) Assertion is true, reason is true, reason is a correct explanation for assertion.

(b) Assertion is true,reason is true, reason is not a correct explanation for assertion.

(c) Assertion is true, reason is false.

(d) Assertion is false, reason is true.

SECTION - B

Section B consists of 5 questions of 2 marks each.

21. If a line has direction ratios $(2, -1, -2)$, then what are its direction cosines?

22. Prove that if E and F are independent events, then the events E and F' are also independent.

23. The total cost $C(x)$ associated with the production of x units of an item is given by $C(x) = 0.005x^3 - 0.02x^2 + 30x + 5000$.

Find the marginal cost when 3 units are produced, where by marginal cost we mean the instantaneous rate of change of total cost at any level of output.

24. Find the general solution of the differential equation $\sec^2 x \tan y\, dx + \sec^2 y \tan x\, dy = 0$

OR

Find the general solution of the differential equation $\frac{dy}{dx} = e^{x+y}$.

25. Minimize $Z = 3x + 5y$ such that $x + 3y \geq 3$, $x + y \geq 2$, x, $y \geq 0$.

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SECTION - C

Section C consists of 6 questions of 3 marks each.

26. Find a matrix A such that $2A-3B+5C=O$,
where $B=\begin{bmatrix}-2 & 2 & 0\\ 3 & 1 & 4\end{bmatrix}$ and $C=\begin{bmatrix}2 & 0 & -2\\ 7 & 1 & 6\end{bmatrix}$.

OR

If $A=\begin{bmatrix}2 & 0 & 1\\ 2 & 1 & 3\\ 1 & -1 & 0\end{bmatrix}$, then find the values of (A^2-5A).

27. Solve graphically the following system of in equations.
$x+2y\geq 20\,,\ 3x+y\leq 15$

28. Find $\int \frac{2\cos x}{(1-\sin x)(2-\cos^2 x)}\,dx$.

29. Find the value of
$\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right)+\cot^{-1}\left(\frac{1}{\sqrt{3}}\right)+\tan^{-1}\left[\sin\left(\frac{-\pi}{2}\right)\right]$.

30. Find the vector and Cartesian equations of the line through the point $(1,2,-4)$ and perpendicular to the lines

$$\vec{r}\ =(8\hat{i}-19\hat{j}+10\hat{k})+\lambda(3\hat{i}-16\hat{j}+7\hat{k})$$

and $$\vec{r}\ =(15\hat{i}+29\hat{j}+5\hat{k})+\mu(3\hat{i}+8\hat{j}-5\hat{k}).$$

31. A ladder 5 m is leaning against a wall .The bottom of the ladder is pulled along the ground anyway from the wall, at the rate 2 cm/s. How fast is its height on the wall decreasing, when the foot of the ladder is 4 meter away from the wall?

OR

A balloon, which always remains spherical on inflation, is being inflated by pumping is 900 cubic centimetres of gas per second. Find the rate at which the radius of the balloon increases when the radius is 15 cm.

SECTION - D

Section D consists of 4 questions of 5 marks each.

32. If $y=x^x$, then prove that $\frac{d^2y}{dx^2}-\frac{1}{y}\left(\frac{dy}{dx}\right)^2-\frac{y}{x}=0$.

OR

For what value of λ, the function defined by $f(x)=\begin{cases}\lambda(x^2+2), & \text{if } x\leq 0\\ 4x+6, & x>0\end{cases}$ is continuous at $x=0$?
Hence, check the differentiability of $f(x)$ at $x=0$.

33. Let $A=\{x\in Z:0\leq x\leq 12\}$. Show that $R=\{(a,b):a,b\in A,|a-b| \text{ is divisible by } 4\}$ is an equivalence relation. Find the set of all elements related to 1. Also write the equivalence class [2].

OR

Let $A=R-\{2\}$, $B=R-\{1\}$. If $f:A\to B$ is a function defined by $f(x)=\left(\frac{x-1}{x-2}\right)$, show that f is one-one and onto. Hence find f^{-1}

34. The scalar product of the vector $\vec{a}=\hat{i}+\hat{j}+\hat{k}$ with a unit vector along the sum of vectors $\vec{b}=2\hat{i}+4\hat{j}-5\hat{k}$ and $\vec{c}=\lambda\hat{i}+2\hat{j}+3\hat{k}$ is equal to 1. Find the value of λ and hence find unit vector along $\vec{b}\times\vec{c}$.

OR

If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are three mutually perpendicular vectors of the same magnitude, then prove that $\vec{a}+\vec{b}+\vec{c}$ is equally inclined with the vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$.

35. Using integration, find the area of the following region.
$\{(x,y)\ :\ |x-1|\leq y\leq\sqrt{5-x^2}\}$

OR

Find the area of the smaller region bounded by the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ and the line $\frac{x}{3}+\frac{y}{2}=1$.
Using integration, find the area of the following region. $\left\{(x,y)\ :\ \frac{x^2}{9}+\frac{y^2}{4}\leq 1\leq\frac{x}{3}+\frac{y}{2}\right\}$

SECTION - E

Case study based questions are compulsory.

36. When it comes to taxes, there are two types of taxes in India - Direct and Indirect tax. The direct tax includes income tax, gift tax, capital gain tax, etc

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while indirect tax includes goods and service tax i.e. GST and any local tax.



A company earns before tax profits of ₹100000. It is committed to making a donation to the Red Cross 10% of its after-tax profits. The Central Government levies income taxes of 50% of profits after deducing charitable donations and any local taxes. The company must also pay local taxes of 10% of its profit less the donation to the Red Cross.

(i) Taking the suitable variable form the system of equation that represent given problem.

(ii) Compute how much the company pays in income taxes, local taxes and as a donation to the Red Cross, using Creamer's Rule.

37. Chemical reaction, a process in which one or more substances, the reactants, are converted to one or more different substances, the products. Substances are either chemical elements or compounds. A chemical reaction rearranges the constituent atoms of the reactants to create different substances as products.



In a certain chemical reaction, a substance is converted into another substance at a rate proportional to the square of the amount of the first substance present at any time t. Initially $(t = 0)$ 50 g of the first substance was present; 1 hr later, only 10 g of it remained.

(i) Find an expression that gives the amount of the first substance present at any time t.

(ii) What is the amount present after 2 hr?

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38. Joint Entrance Examination – Advanced, is an academic examination held annually in India. It is organised by one of the seven zonal IITs under the guidance of the Joint Admission Board on a round-robin rotation pattern for the qualifying candidates of the JEE-Main. A candidate can attempt JEE (Advanced) maximum of two times in two consecutive years. A successful candidates get the admission in any IITs of India based on merit.



Applicants have a 0.26 probability of passing IIT advanced test when they take it for the first time, and if they pass it they can get admission in IIT. However, if they fail the test the first time, they must take the test a second time, and when applicants take the test for the second time there is a 0.43 chance that they will pass and be allowed to get admission. Applicants are rejected if the test is failed on the second attempt.

(i) What is the probability that an applicant gets admission in IIT but needs two attempts at the test?

(ii) What is the probability that an applicant gets admission in IIT?

(iii) If an applicant gets admission in IIT, what is the probability that he or she passed the test on the first attempt?

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